

```
#Expected: Should print Same object (since both lists contain equal values)

list1 = [1, 2, 3]

list2 = [1, 2, 3]

if list1 is list2:
    print("Same object")
else:
    print("Different objects")
```

Different objects

```
#Expected: With has_id = False, should print Cannot enter
age = 25

has_id = False

if age >= 18 and has_id:
    print("Can enter")
else:
    print("Cannot enter")
```

Cannot enter

```
#ExpectedShould print Total: 60 (the sum of all three prices)
prices = [10, 20, 30]

total = 0

for price in prices:
    total += price

print("Total:", total)
```

Total: 60

```
#Expected:Should print 10 (the highest score)
scores = ["9", "10", "2"]
print(max(scores, key=int))
```

10

```
#Expected: With is_weekday = True and is_holiday = True, should print No work t
is_weekday = True
```

```
is_holiday = True

if is_weekday and is_holiday:
    print("No work today")
else:
    print("Work today")
```

No work today

```
#Expected``age = 15` should print `Minor`
age = 15

status = "Minor" if age < 18 else "Adult"

print(status)
```

Minor

```
#Expected: classify(50)` should print `Small positive
def classify(num):
    if num > 0:
        if num > 100:
            print("Large positive")
        else:
            print("Small positive")
    elif num < 0:
        print("Negative")
    else:
        print("Zero")

classify(50)
```

Small positive

```
#Expected:check_grade(85) should print B
def check_grade(score):
    match score:
        case score if score >= 90:
            print("A")
        case score if score >= 80:
            print("B")
        case _:
            print("F")

check_grade(85)
```

B

```
#Expected: check_number(-5) should print `Negative`
def check_number(num):
    if num > 0:
        print("Positive")
    elif num < 0:
        print("Negative")
    else:
        print("Zero")

check_number(-5)
```

Negative

```
#Expected: With items = [1, 2, 3], should print `List has items`
items = [1, 2, 3]
if items:
    print("List has items")
else:
    print("List is empty")
```

List has items

```
#Expected: Should print Original fruits list: ['apple'] (unchanged)
```

```
def add_item(item, my_list):
    updated_list = my_list.copy()
    updated_list.append(item)
    return updated_list

fruits = ["apple"]

new_list = add_item("banana", fruits)

print("Original fruits list:", fruits)
print("New list:", new_list)
```

Original fruits list: ['apple']
New list: ['apple', 'banana']

```
#Expected: Should print Total after discount: 80
total = 100
```

```
def apply_discount(total, discount):
    total = total - discount
    return total

total = apply_discount(total, 20)

print("Total after discount:", total)
```

Total after discount: 80

#Expected: Should print 5 4 3 2 1 Done!

```
def countdown(n):  
    if n == 0:  
        print("Done!")  
    else:  
        print(n, end=" ")  
        countdown(n - 1)  
  
countdown(5)
```

5 4 3 2 1 Done!

#Expected: calculate_total(100) should print 110.0

```
def calculate_total(price, tax=None):  
    if tax is None:  
        tax = price * 0.1  
    return price + tax  
  
print(calculate_total(100,20))
```

120

#Expected: Should print Alice

```
def print_details(**kwargs):  
    print(kwargs["name"])  
  
print_details(name="Alice", age=25)
```

Alice

#Expected: Should print Found greeting

```
text = "Hello World"  
if "hello" in text.lower():  
    print("Found greeting")  
else:  
    print("Not found")
```

Found greeting

Expected: Should print [[1, 0, 0], [0, 0, 0], [0, 0, 0]]

```
grid = [[0] * 3 for _ in range(3)]  
grid[0][0] = 1  
print(grid)
```

```
[[1, 0, 0], [0, 0, 0], [0, 0, 0]]
```

```
#Expected: Should print (1, 2, 3, 4, 5)
coordinates = (1, 2, 3)
coordinates = coordinates + (4, 5)
print(coordinates)
```

```
(1, 2, 3, 4, 5)
```

```
#Expected: Should print the elements unique to set a: `{1, 2}`
a = {1, 2, 3, 4}
b = {3, 4, 5, 6}
unique_to_a = a - b
print(unique_to_a)
```

```
{1, 2}
```

```
# Expected:Should print `{ 'apples': [10], 'bananas': [], 'cherries': []}`
inventory = {key: [] for key in ["apples", "bananas", "cherries"]}
inventory["apples"].append(10)
print(inventory)
```

```
{'apples': [10], 'bananas': [], 'cherries': []}
```

```
#Expected: Should print `4` (-2 squared)
result = (-2) ** 2
print(result)
```

```
4
```

```
#Expected:With fruit = "cherry", should print `Different fruit`
fruit = "cherry"
if fruit in ("apple","banana"):
    print("It's apple or banana")
else:
    print("Different fruit")
```

```
Different fruit
```

```
#Expected: Should print `Square is: 25`
def square(x):
    return x * x

value = square(5)
print("Square is:", value)
```

```
Square is: 25
```

```
#Should print [1,2,5,8]
numbers = [5, 2, 8, 1]
sorted_numbers = sorted(numbers)
print(sorted_numbers)
```

```
[1, 2, 5, 8]
```

```
#Expected:Should print Python-is-fun
words = ["Python", "is", "fun"]
sentence = "-".join(words)
print(sentence)
```

```
Python-is-fun
```